Competitive Programming

Week 1 : Assignment 2 :

Batch - 06
2303A51399
Assignment 2 :

-->Q. Shelf Product Lookup (Divide and Conquer)
Problem Statement
A supermarket maintains a sorted list of product IDs. For each query ID X, determine
how many times X appears in the list. You must solve using binary-search style divide
and conquer (find first occurrence and last occurrence).
Input Format The first line contains integer T. For each test case:
- First line: N Q
- Second line: N integers (sorted product IDs)
- Next Q lines: integer X (query ID)
Output Format For each query, print the count of X on a new line.
Constraints
- $1 \leqslant T \leqslant 20$
- $1 \leqslant N \leqslant 200000$ (sum of N over all test cases $\leqslant 200000$)
- $1 \leqslant Q \leqslant 200000$ (sum of Q over all test cases $\leqslant 200000$)
- Product IDs fit in 32-bit signed integer
Sample Input
1
8 3
101 101 102 102 102 105 110 110 101 102 999
Expected Output
230

-->CODE

```
def first_occurrence(arr, x):
    low = 0
    high = Len(arr) - 1
    ans = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == x:
            ans = mid
            high = mid - 1
```

```
        elif arr[mid] < x:
            low = mid + 1
        else:
            high = mid - 1

    return ans


def last_occurrence(arr, x):
    low = 0
    high = Len(arr) - 1
    ans = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == x:
            ans = mid
            low = mid + 1
        elif arr[mid] < x:
            low = mid + 1
        else:
            high = mid - 1

    return ans


t = int(input())

for _ in range(t):
    n, q = map(int, input().split())

    nums = list(map(int, input().split()))

    arr = numbs[:n]
    queries = nums[n:]

    output = ""

    for x in queries:
        first = first_occurrence(arr, x)

        if first == -1:
            output += "0"
```

```
        else:
            last = last_occurrence(arr, x)
            output += str(last - first + 1)

    print(output)
```

-->Input:

1
8 3
101 101 102 102 102 105 110 110 101 102 999
230